

Planetary Classification in Amateur Astronomy

IMRAN SULTAN

Abstract

In this paper we develop two methods to classify images of the gas giants Jupiter and Saturn. We create training and testing data sets using images captured and processed by amateur astronomers. For our color classification method, we represent images by their most prominent colors. We find that the two-color Fisher linear discriminant ($t_c = -2.19$) is most successful in maximizing the true positive rate while minimizing the false positive rate in our training data sets. For our shape classification method, we characterize the shapes of planets by carrying out a least squares elliptical fitting procedure. We find that eccentricity is an excellent discriminant ($e_c = 0.75$) and produces significant spacing between the distributions of the training data sets. We conclude that classification by shape is a significantly more accurate method compared to classification by color. The accuracies of our classification methods when applied to the testing data sets are 99% for the shape classifier, and 87% for the two-color classifier.

1. INTRODUCTION

Due to advancements in consumer telescopes and other optical equipment, as well as the release of free publicly available image processing software, amateur astronomy and astrophotography are becoming increasingly accessible hobbies. From backyards, amateur astronomers can now capture high-quality images of astronomical objects both within and outside our Solar System, on a non-astronomical budget. Among the most common objects photographed are the local planets, including the gas giants Jupiter and Saturn. These planets are simple to find in the night sky, even in areas with high light pollution.

Figure 1 shows an example workflow for processing an image of Saturn. (a) First, a video of Saturn containing ~ 5000 frames was captured with the SharpCap¹ astrophotography software. The optical equipment used were an Orion Skyquest XT8 eight-inch Dobsonian telescope, Celestron Omni 2x Barlow Lens, and ZWO ASI120MC-S camera. The raw video file was pre-processed with PIPP², which centered Saturn in each frame. (b) The best $\sim 30\%$ of frames were selected and combined to form a single image through ‘image stacking’ using the AutoStakkert!³ lucky imaging software. The output image was also sharpened. (c) For color correction, the image was processed with RegiStax 6⁴. The RGB alignment, RGB balance, histogram, and wavelets tools were used. (d) Finally, the contrast and saturation of the image were adjusted in GIMP⁵.

As illustrated by the workflow, there are many user-controlled degrees of freedom throughout the process from a raw video of a planet to the final image (e.g., the amount of

frames used for image stacking, and the parameters associated with RGB color correction). The upshot is that there are significant differences in the quality and processing of planetary images, which can make the classification of planets nontrivial; this variance is amplified by the fact that amateur astronomers typically favor aesthetics over maximizing the scientific content of the final image (in this sense, amateur astronomy is more of an art than a science).

In this paper, we will develop, test, and compare planetary classification methods. Our central goal will be an algorithm that takes an image of a planet as its input, and accurately labels it as either Jupiter or Saturn. In Section 2, we list our image data sets and describe the initial processing of the images. We classify planets by the prominent colors in an image and their shapes in Sections 3 and 4, respectively. Finally, in Section 5 we summarize and discuss our results.

2. DATA SETS

In this section, we list the data sets used in our analysis, and describe our initial data processing. [r/astrophotography](https://www.reddit.com/r/astrophotography) is an online amateur astrophotography community hosted on the social networking website Reddit. The community has close to two million subscribers, and users frequently submit planetary images they have captured. We use the submissions to create four data sets of Jupiter and Saturn. As users around the world have captured and processed the images using a variety of optical equipment (telescopes, lenses, and cameras) and image processing programs, we expect significant differences in the data.

We first store hundreds of images of Jupiter and Saturn that [r/astrophotography](https://www.reddit.com/r/astrophotography) users have submitted. We carry out an initial stage of processing: we crop the images to remove any extra astronomical objects (such as moons) as well as text labels. We then use an algorithm adapted from [nathancy \(2020\)](#) to detect the planet in each image and tightly crop the image. A subset of the resulting images are shown in Figure 2.

¹ <https://www.sharpcap.co.uk>

² <https://sites.google.com/site/astropipp>

³ <https://www.autostakkert.com>

⁴ <https://www.astronomie.be/registax>

⁵ <https://www.gimp.org>

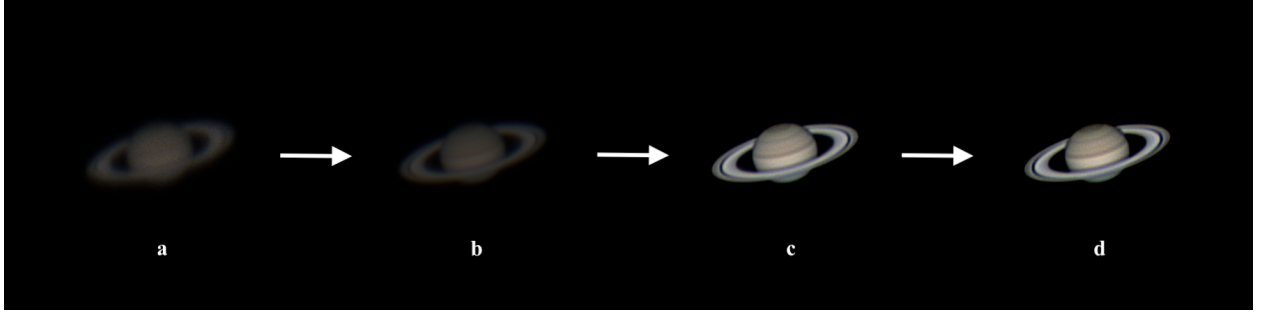


Figure 1. Example workflow for processing a planetary image. The major steps are outlined in the four panels. (a) A video of Saturn with ~ 5000 frames was captured (using SharpCap) and pre-processed (using PIPP). (b) To remove atmospheric noise, the best $\sim 30\%$ of frames were combined to form a single image through ‘image stacking,’ and the resulting image was sharpened (in AutoStakkert!). (c) Color correction was implemented, principally by using the wavelets tool of RegiStax 6. (d) Slight adjustments to the contrast and saturation of the image were made in GIMP.

Table 1. Data sets of planetary images

Data set	Number of images
jupiter-training	85
saturn-training	85
jupiter-testing	85
saturn-testing	85

The resulting data sets are listed in Table 1. In particular, we divide the images of Jupiter and Saturn into two data sets each, which we refer to as the ‘training’ and ‘testing’ data sets. In the following sections, we will use the training data sets to develop our classifiers and select parameter values. Once we have created the classifiers, we will use the testing data as a benchmark for the classifiers. The primary metric we will use when evaluating the testing data will be the *accuracy*, which we define as the ratio of the number of correct classifications across both jupiter-testing and saturn-testing, to the 170 total elements in the two testing data sets.

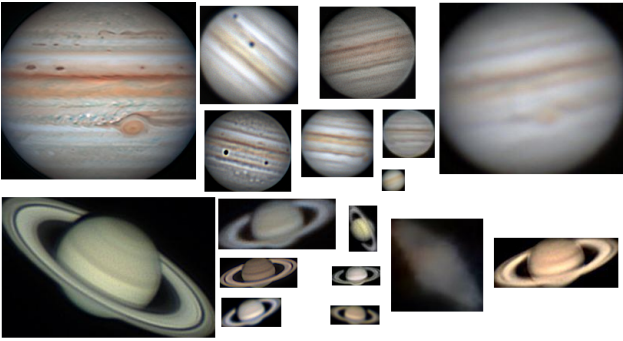


Figure 2. A subset of images in the jupiter-training and saturn-training data sets. There are significant differences in the sharpness, color, orientation, and size of the planets for both data sets; a good classifier must be able to label planets despite the variance within the data. All images were submitted by users of the *r/astrophotography* online community.

3. CLASSIFICATION BY COLOR

Jupiter and Saturn have different color profiles, which are, in principle, reflected in processed images of the planets. In this section we take advantage of the difference to classify the planets in our data sets by their representative colors.

3.1. Representing images as color vectors

First we devise a method to associate every image in our data sets with a *color* that is representative of the image. We define a color as a list of three numbers between 0 and 255 corresponding to the red (R), green (G), and blue (B) channels. Possible ways to do this include computing the mean, median, or mode of the RGB pixel values of an image.

We generalize our problem such that for a given image, we associate n colors that are representative of the image. Thus, we represent each element of our data sets as a vector $\vec{x}_n$ in a $3n$ -dimensional space,

$$\vec{x}_n = (R_1, G_1, B_1, \dots, R_n, G_n, B_n) \quad (1)$$

To find $\vec{x}_n$, we use the software *Color Thief*⁶ which generates a color palette (of n colors) associated with an image by median cut.

3.2. Fisher linear discriminant

After we have calculated a color vector $\vec{x}_n$ for each image, we need a method to cluster the points in $3n$ -space corresponding to each planet, and classify new points as either Jupiter or Saturn. We reduce the $3n$ -dimensional clustering problem to one dimension using the Fisher linear discriminant t_n . In this analysis, we use our training data sets (see Table 1) to develop an expression for t_n .

To calculate t_n given an image and value of n , we define

$$W_n = V_{n, \text{Jupiter}} + V_{n, \text{Saturn}} \quad (2)$$

$$\vec{a}_n = W_n^{-1} \Delta \vec{x}_n \quad (3)$$

$$(\Delta \vec{x}_n)_i = (\vec{x}_{n, \text{Saturn}})_i - (\vec{x}_{n, \text{Jupiter}})_i \quad (4)$$

⁶ <https://github.com/fengsp/color-thief-py>

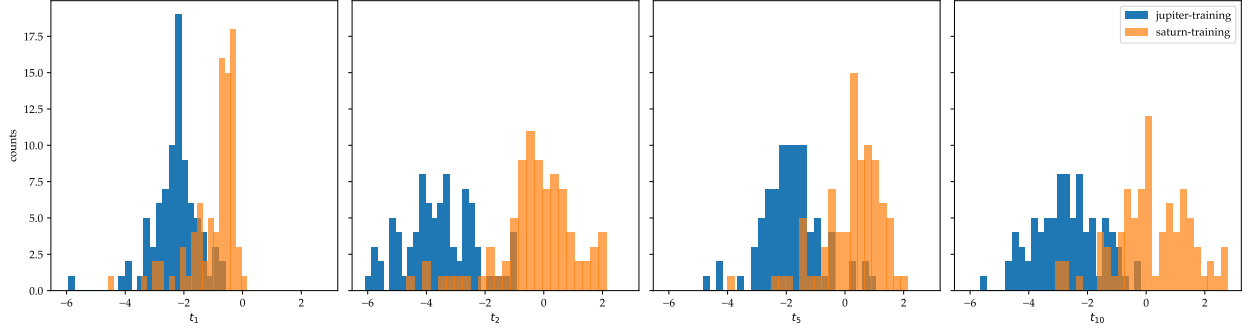


Figure 3. Distributions of the Fisher linear discriminant t_n calculated for the jupiter-training (blue) and saturn-training (orange) data sets. The panels show the linear discriminant calculated from color vectors of one, two, five, and ten representative colors.

where the $3n \times 3n$ covariance matrices V_n include the variances and covariances in the elements of the vectors $\vec{x}_n$ for each planet's data set. $(\vec{x}_n)_i$ is the mean of the i th element of the color vector in each data set. The linear discriminant is then given by

$$t_n(\vec{x}_n) = \vec{a}_n \cdot \vec{x}_n \quad (5)$$

3.3. Results

In order to determine the optimal number of colors to use in the linear discriminant for the classification, we explore the range $n = 1$ to $n = 10$. The resulting t_n distributions found using the Jupiter and Saturn training data sets are shown in Figure 3 for $n = 1, 2, 5$, and 10 . Out of the four panels, the two color and ten color discriminants appear especially promising to use for classification, as overlap between the distributions of the two planets is relatively low.

To formalize the selection of the optimal value of n , we utilize receiver operating characteristic (ROC) curves (see, e.g., [Gneiting and Vogel 2018](#)). For a given threshold t_c , we find the true positive rate (TPR) in the saturn-training data set and the false positive rate (FPR) in the jupiter-training data set. The resulting ROC curves for the 10 n values are shown in Figure 4, calculated over $t_c \in [-5, 5]$.

Using the ROC curves, we select the optimal threshold value t_c to use as our classifier by noting that the best discriminant should maximize the TPR and minimize the FPR. For each ROC curve, we find the value t_c for which the curve is closest to the point $(0, 1)$. The resulting optimal threshold values t_c are listed in Table 2 for $1 \leq n \leq 10$. We also provide the Euclidean distance of the curve to $(0, 1)$ at t_c , which we call Δ .

The best discriminants are $n = 2$ and $n = 10$, as these minimize Δ . This is consistent with our observation in Figure 3. Overall, the best discriminant is $t_c = -2.19$ for $n = 2$. Further tests using additional data sets will be required to determine whether the two-color space is inherently the best classifier, or if this is simply an artifact of the particular images in our training data sets.

Finally, we apply our discriminants (with the chosen threshold values of Table 2) to our testing data sets. The accuracy (defined in Section 2) is given in the last column

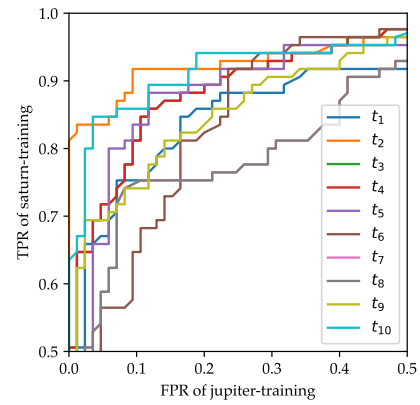


Figure 4. ROC curves shown for $1 \leq n \leq 10$ and $-5 \leq t_c \leq 5$. Plotted is the true positive rate (TPR) in the saturn-training data set and the false positive rate (FPR) in the jupiter-training data set. The $n = 2$ and $n = 10$ curves best maximize the TPR while minimizing the FPR.

Table 2. Linear discriminant results for classification using n colors

n	t_c	Δ	Accuracy (%)
1	-1.64	0.225	72.4
2	-2.19	0.125	86.5
3	-1.21	0.184	85.3
4	-1.21	0.184	85.3
5	-0.92	0.166	80.0
6	-0.09	0.249	78.2
7	-0.11	0.269	78.2
8	-0.11	0.269	78.2
9	-0.73	0.235	79.4
10	-1.01	0.157	84.1

of the table for each n . We achieve a maximum accuracy of 86.5% for $n = 2$. As $n = 2$ also minimized Δ for the training data, this indicates that our training and testing data sets may be statistically similar, which should ideally be the case for our analysis.

4. CLASSIFICATION BY SHAPE

In addition to having different color profiles, Jupiter and Saturn also have distinct shapes (due in part to Saturn’s prominent rings). In this section we develop a method to characterize the shape of a photographed planet, and apply the method to classify the planets in our data sets.

4.1. Using eccentricity as a discriminant

To characterize the shape of a planet, we note that Jupiter typically occupies a circular region in images. On the other hand, the space occupied by Saturn is stretched along an axis due to its rings, and the region appears elliptical. Thus, we must differentiate between a circle and an ellipse to classify our data sets. We use the eccentricity e as our discriminant. Given an ellipse of semi-major axis a and semi-minor axis b (where $a = b$ for a circle), the eccentricity is defined as

$$e = \sqrt{1 - (b/a)^2} \quad (6)$$

Our method for finding the eccentricity of an image is illustrated in Figure 6. Here, we describe the four main steps.

- (a) We represent the image in a two-dimension coordinate grid. The planet in the image can have an arbitrary angle of tilt with respect to the axes of this coordinate system.
- (b) As we carried out in Section 2, we apply an algorithm adapted from [nathancy \(2020\)](#) to detect the planet in

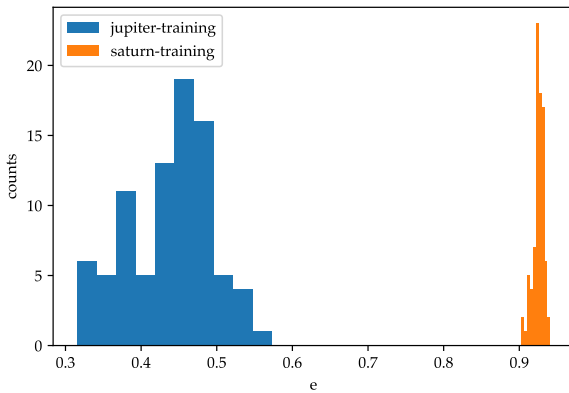


Figure 5. Distributions of the eccentricity e calculated for the jupiter-training (blue) and saturn-training (orange) data sets. As we expect after noting the roughly circular shape of Jupiter in images, the eccentricities for the jupiter-training set are closer to zero (for a circle, $e = 0$). The eccentricity is an excellent discriminant, with a large gap between the two distributions.

the image. Points on the pixel grid that include the planet are assigned a value of 1 (yellow), and background points are assigned a value of 0 (purple).

- (c) Using the `scipy` library, we find the convex hull of the points with a value of 1. The result is a set of points which represent the boundary of the planet in the image (blue points).
- (d) We fit the equation of an ellipse to the vertices of the convex hull using the least squares elliptic fitting software of [Hammel and Sullivan-Molina \(2020\)](#). We use the a and b values found in the elliptical fitting to calculate e by equation 6.

4.2. Results

We first apply our algorithm to the images in the training data sets. Figure 5 shows the resulting eccentricity distributions for the jupiter-training (blue) and saturn-training (orange) data sets. The eccentricity is an excellent discriminant, with a large gap between the two distributions and no overlap. We select $e_c = 0.75$ as the threshold for classification of Jupiter ($e < e_c$) and Saturn ($e > e_c$).

Finally, we apply the classifier to the testing data sets. For the jupiter-testing and saturn-testing data sets, the classifier has a nearly-perfect accuracy of 99.4%. Compared to our results in Section 3.3 for the color classifier, classification by shape is a significantly more accurate method of distinguishing Jupiter and Saturn; we discuss possible reasons for this in the next section.

5. SUMMARY AND DISCUSSION

In this paper we developed and tested two methods to classify images of the most massive planets of the Solar System, the gas giants Jupiter and Saturn. We created training and testing data sets for Jupiter and Saturn using images captured and processed by amateur astronomers; the images were submitted to a popular online astrophotography community.

For our color classification method, we represented each image of our data sets by its n most prominent colors (where each color has three channels). We used the Fisher linear discriminant to reduce the $3n$ -dimensional clustering problem to one dimension. We found that the two-color classifier with $t_c = -2.19$ was most successful in maximizing the true positive rate while minimizing the false positive rate in our training data sets.

For our shape classification method, we characterized the shape of the region occupied by the planet in each image (circular for Jupiter and elliptical for Saturn). We used the eccentricity as our discriminant, which we found by fitting the equation of an ellipse to the boundary points of the foreground of each image in our data sets; we used a least squares fitting procedure. We found that the eccentricity is an extremely effective discriminant, as there was significant spacing between the e -distributions of our two training data sets. We selected $e_c = 0.75$ as the threshold for classification.

We found that classification by shape is a significantly more accurate method of distinguishing Jupiter and Saturn

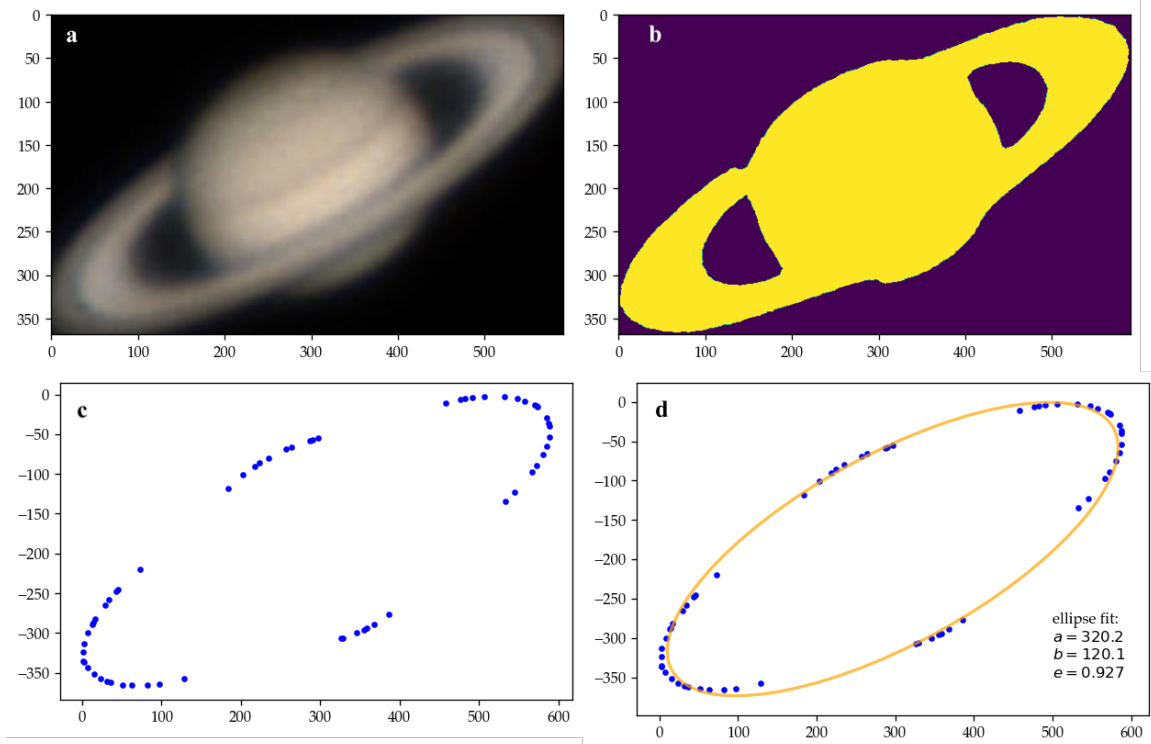


Figure 6. Procedure to calculate the eccentricity of Saturn in an image from the saturn-training data set, shown in panel a. (b) We distinguish pixels of the planet (yellow) from the background (purple). (c) We find the convex hull of the foreground points. The blue points indicate the vertices of the convex hull. (d) We fit the equation of an ellipse to the blue points with a least squares algorithm. The resulting values of the semi-major axis, semi-minor axis, and eccentricity are indicated in the panel.

compared to classification by color. The accuracies of our classification methods when applied to the testing data sets were 99.4% for the shape classifier, and 86.5% for the two-color classifier. This may be a result of the numerous user-controlled degrees of freedom related to RGB color correc-

tion in planetary image processing. Throughout the image editing process (outlined in Figure 1), the shape of the planet remains approximately constant while colors are progressively adjusted. Thus, there can be a large variance in the colors of images processed by different amateur astronomers, while the shapes of planets is uniform.

REFERENCES

Gneiting, T. and Vogel, P. 2018, arXiv:1809.04808 [stat.ME]

Hammel, B. and Sullivan-Molina, N. 2020, bdhammel/least-squares-ellipse-fitting: v2.0.0 (Version v2.0.0), Zenodo
nathancy, 2020, <https://stackoverflow.com/questions/60033008/remove-image-background-and-extract-the-object-in-image/60050200#60050200>